

List

1. Redis List

List는 문자열 값을 순서대로 저장하는 자료구조입니다. 왼쪽과 오른쪽에서 값을 빠르게 넣거나 뺄 수 있어 최근 기록, 대기열, 간단한 큐에 적합합니다.

왼쪽 <- [A] [B] [C] -> 오른쪽

2. 값 추가와 조회

- LPUSH : 왼쪽에 추가

```
DEL demo:list  
LPUSH demo:list B  
LPUSH demo:list A
```



Databases > My Redis Stack Database db0  

0.32 % | 0 | 2 MB | 1 | 4  Insights

 1 DEL demo:list
 2 LPUSH demo:list B
 3 LPUSH demo:list A






Clear Results

LPUSH demo:list A	Jun 22, 15:22:35	3.609 msec				
(integer) 2						
LPUSH demo:list B	Jun 22, 15:22:35	3.734 msec				
(integer) 1						
DEL demo:list	Jun 22, 15:22:35	1.027 msec				
(integer) 0						

browser에서 저장된 데이터 확인

The screenshot displays the Redis Stack browser interface. The top navigation bar shows the path "Databases > My Redis Stack Database" and the selected database "db0". On the right, system metrics are displayed: 0.26 % memory usage, 0 connections, 2 MB memory used, 1 key, and 4 users. The left sidebar contains navigation icons for keys, documents, graphs, and connections.

The main content area is divided into two panels. The left panel shows a list of keys with the following details:

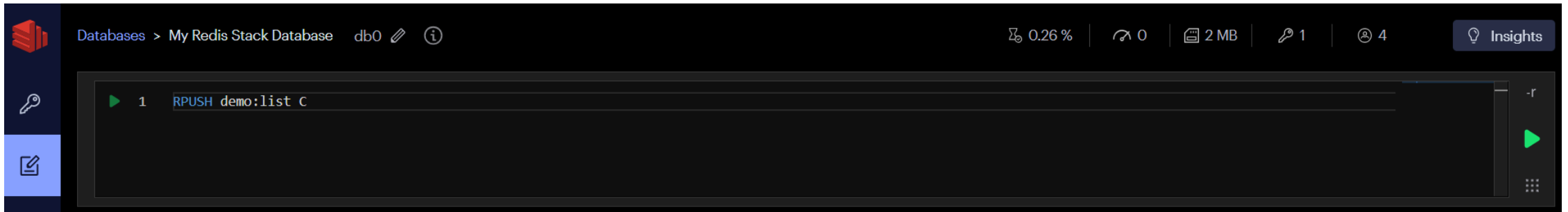
Key Type	Key Name	Limit	Size
LIST	demo:list	No limit	72 B

The right panel provides detailed information for the selected key "demo:list". It shows the key type as "LIST", a size of 72 B, a length of 2, and a TTL of "No limit". Below this, a table lists the elements in the list:

Index	Element
0	A
1	B

- **RPUSH** : 오른쪽에 추가

```
RPUSH demo:list C
```



Clear Results			
RPUSH demo:list C	Jun 22, 15:26:24	0.787 msec	   
(integer) 3			
LPUSH demo:list A	Jun 22, 15:24:55	0.915 msec	  

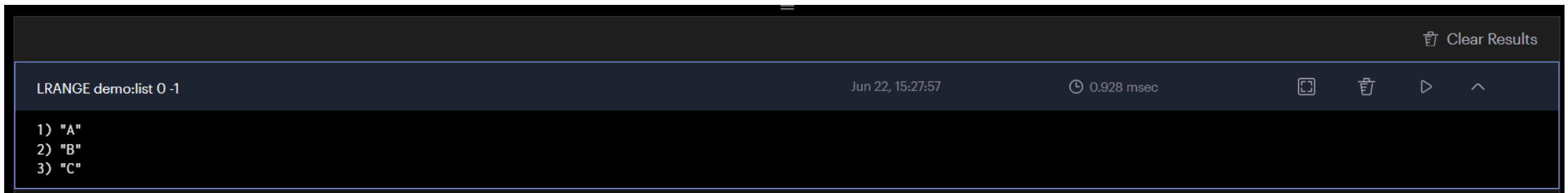
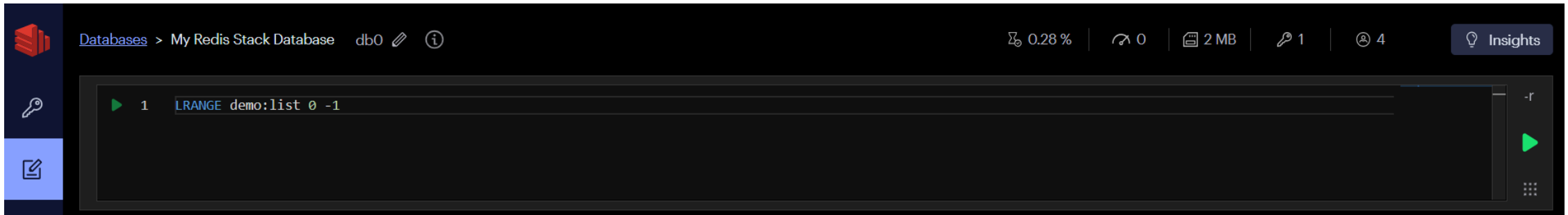
- `LRANGE key start stop` : 범위 조회

인덱스 `0` 은 첫 번째 값이고 `-1` 은 마지막 값입니다.

따라서 `LRANGE key 0 -1` 은 전체 목록을 조회합니다.

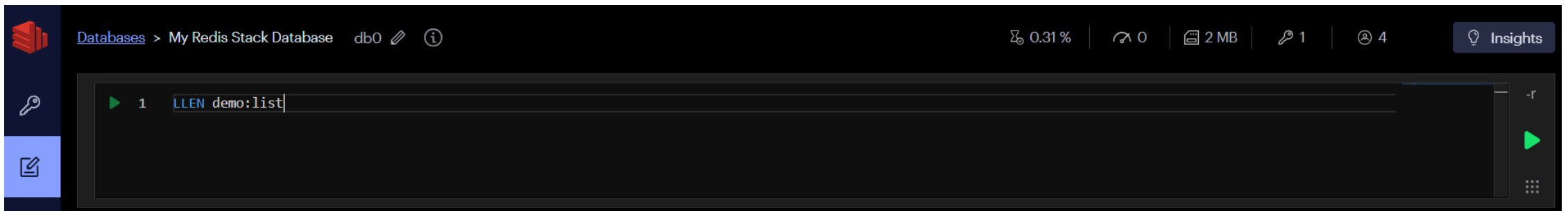
운영 환경에서는 목록이 매우 길 수 있으므로 전체 대신 필요한 범위만 조회합니다.

```
LRANGE demo:list 0 -1
```

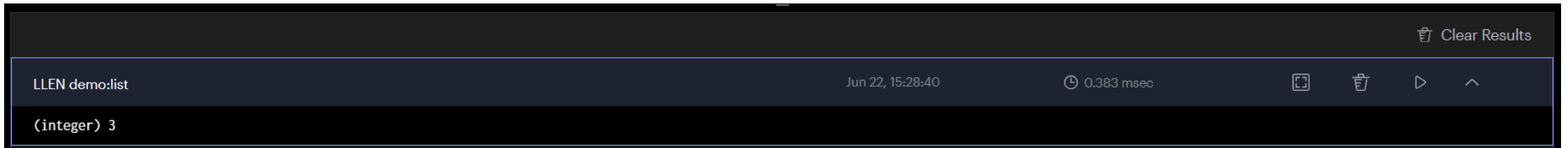


- LLEN : 목록 길이

```
LLEN demo:list
```



The screenshot shows the Redis Stack interface. The top bar indicates the database is 'My Redis Stack Database' (db0). The query editor contains the command `LLEN demo:list`. The interface includes a sidebar with icons for databases, keys, and documents, and a top bar with various status indicators like memory usage (0.31%), connections (0), and a 'Clear Results' button.



The screenshot shows the result of the `LLEN demo:list` command. The result is displayed as `(integer) 3`. The interface includes a top bar with a 'Clear Results' button and a bottom bar showing the command, timestamp (Jun 22, 15:28:40), and execution time (0.383 msec).

3. 값 꺼내기

LPOP 과 RPOP 은 값을 반환하면서 목록에서 제거합니다.

```
LPOP demo:list  
RPOP demo:list  
LRANGE demo:list 0 -1
```



[Databases](#) > My Redis Stack Database db0  

0.28 % | 0 | 2 MB | 1 | 4  Insights



```
1 LPOP demo:list
2 RPOP demo:list
3 LRange demo:list 0 -1
```

-r



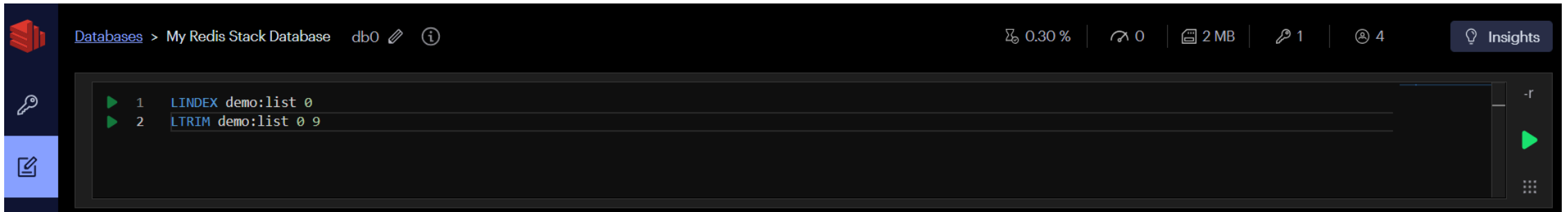


 Clear Results

LRange demo:list 0 -1	Jun 22, 15:30:41	1.116 msec				
1) "B"						
RPOP demo:list	Jun 22, 15:30:41	1.242 msec				
"C"						
LPOP demo:list	Jun 22, 15:30:41	1.443 msec				
"A"						

조회만 하려면 `LINDEX(단일 항목)` 또는 `LRANGE(범위 항목)` 를, 리스트를 특정 범위로 영구적으로 잘라내려면 `LTRIM`을 사용합니다

```
LINDEX demo:list 0  
LTRIM demo:list 0 9
```



4. 최근 조회 상품

최근 본 상품 추가 (새 항목이 앞에 쌓임)

```
LPUSH user:7:recent-products 1001
```

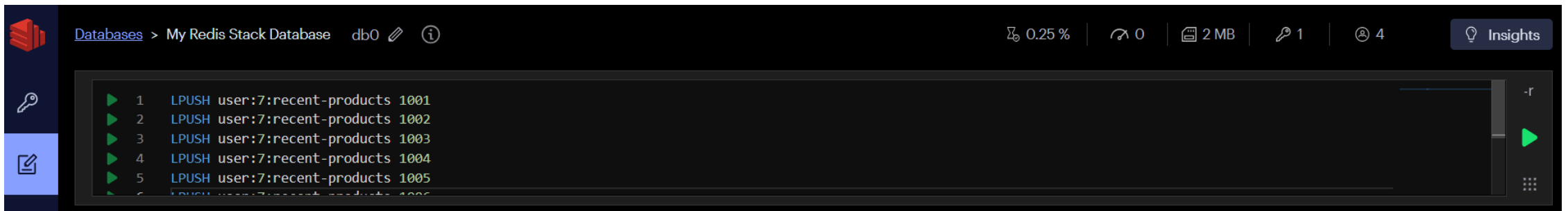
```
LPUSH user:7:recent-products 1002
```

```
LPUSH user:7:recent-products 1003
```

```
LPUSH user:7:recent-products 1004
```

```
LPUSH user:7:recent-products 1005
```

```
LPUSH user:7:recent-products 1006
```



현재 상태: [1006, 1005, 1004, 1003, 1002, 1001]

The screenshot displays the AWS IAM console interface for a Redis Stack database named 'db0'. The left sidebar contains navigation icons for Databases, Keys, Clusters, and a BETA section. The main content area shows the 'Databases' view with a list of keys. Two keys are listed: 'demo:list' (72 B) and 'user:7:recent-products' (96 B). The 'user:7:recent-products' key is selected, and its details are shown on the right. The key is of type 'LIST' and has a length of 6. The elements are listed in a table with indices 0 through 5, corresponding to the values 1006, 1005, 1004, 1003, 1002, and 1001. The interface also includes a search bar, a filter dropdown, and a 'Bulk Actions' button.

Database: db0

Keys:

Key Name	Type	Value	Size
demo:list	LIST	No limit	72 B
user:7:recent-products	LIST	No limit	96 B

Key Details: user:7:recent-products

96 B Length: 6 TTL: No limit

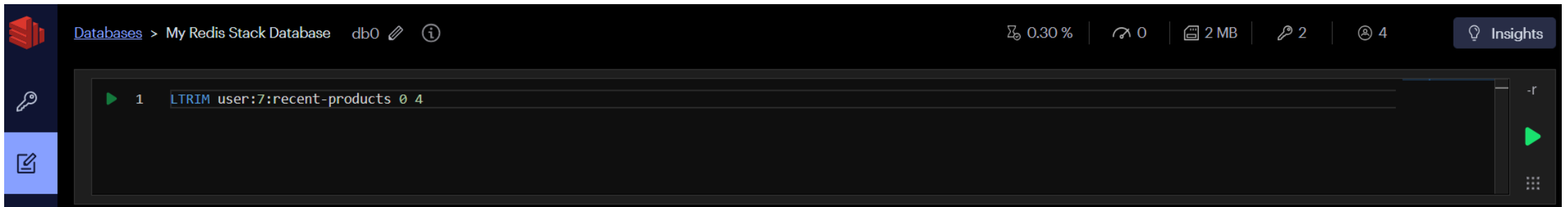
Index	Element
0	1006
1	1005
2	1004
3	1003
4	1002
5	1001

핵심 포인트는 두 가지예요.

- LPUSH 로 새 상품을 앞에 추가 → 최신 순으로 자동 정렬
- LTRIM 0 4 로 5개 초과 시 오래된 항목을 자동 제거

실무에서는 상품을 추가할 때마다 이 두 명령어를 항상 같이 실행합니다.

```
# 최근 5개만 유지 (오래된 항목 1001 자동 제거)  
LTRIM user:7:recent-products 0 4
```



결과 확인: [1006, 1005, 1004, 1003, 1002]

The screenshot shows the Redis Stack database interface. The top navigation bar includes 'Databases > My Redis Stack Database', 'db0', and various status indicators (0.28%, 0, 2 MB, 2, 4, Insights). The left sidebar contains icons for key management, analytics, and a 'BETA' label. The main content area displays a list of keys with the following details:

Key Type	Key Name	Limit	Size
LIST	demo:list	No limit	72 B
LIST	user:7:recent-products	No limit	88 B

The 'user:7:recent-products' key is selected, showing its details: 88 B, Length: 5, TTL: No limit, and a refresh time of < 1 min. The key is a LIST type. The elements of the list are displayed in a table:

Index	Element
0	1006
1	1005
2	1004
3	1003
4	1002

전체 데이터 조회

```
LRANGE user:7:recent-products 0 -1
```

The screenshot displays the Redis Stack database interface. The top navigation bar shows 'Databases > My Redis Stack Database db0' with various status icons (0.29%, 0, 2 MB, 2, 4) and an 'Insights' button. The main query editor contains the command `LRANGE user:7:recent-products 0 -1`. Below the editor, the results are displayed in a list format:

```
LRANGE user:7:recent-products 0 -1
```

Jun 22, 15:46:35 0.807 msec

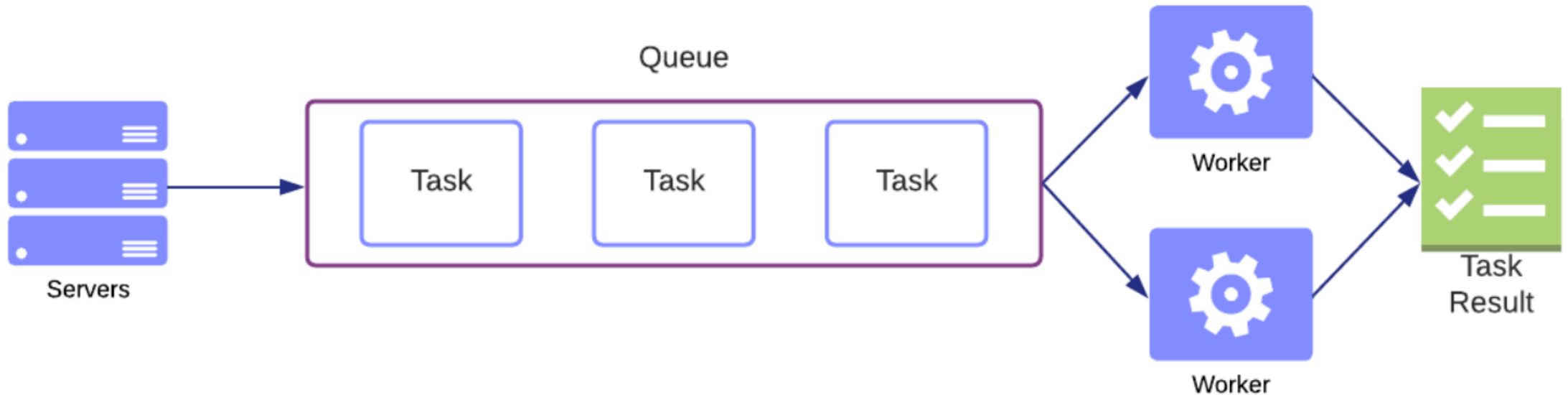
- 1) "1006"
- 2) "1005"
- 3) "1004"
- 4) "1003"
- 5) "1002"

The interface also includes a 'Clear Results' button in the top right corner of the results pane.

5. 작업 큐

생산자는 오른쪽에 작업을 추가하고 소비자는 왼쪽에서 꺼냅니다.

Servers(생산자) → [작업큐] → Worker A(소비자)가 꺼내서 처리 → 완료 표시
↑
Worker B(소비자)는 못 가져감 (이미 누가 가져갔으니)



Redis 큐가 필요한 3가지 이유

1. 속도 차이 흡수

- 요청은 빠르게 들어오는데, 처리는 느릴 때
- 예) 주문 접수(빠름) vs DB 저장 + 알림(느림)

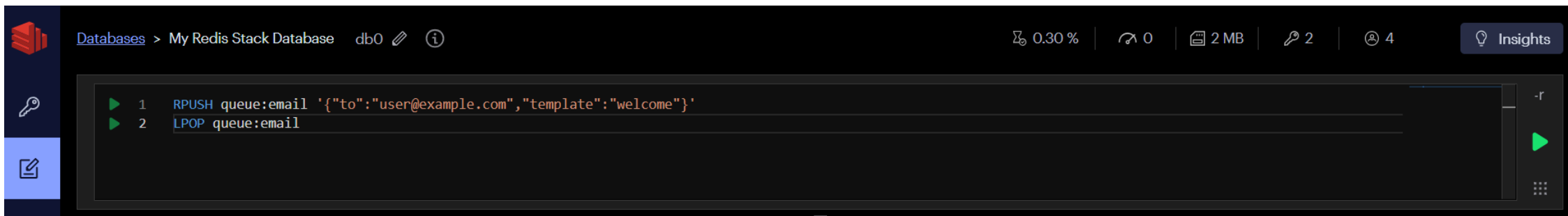
2. 트래픽 폭주 방어

- 갑자기 요청 1000개 와도 큐가 버퍼 역할
- Worker는 자기 속도대로 처리

3. 서버 죽어도 작업 안 사라짐

- 큐에 쌓여 있으니 서버 재시작 후 이어서 처리 가능

```
RPUSH queue:email '{"to":"user@example.com","template":"welcome"}'  
LPOP queue:email
```

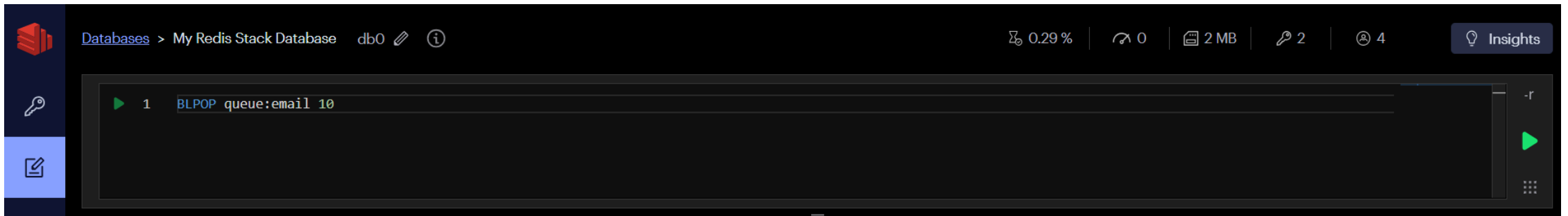


The screenshot shows the Redis Stack interface. The top bar indicates the database is 'db0'. The command history shows two commands: 1. `RPUSH queue:email '{"to":"user@example.com","template":"welcome"}'` and 2. `LPOP queue:email`. The interface includes a sidebar with icons for databases, keys, and documents, and a top bar with performance metrics like 0.30% CPU, 0 connections, 2 MB memory, 2 keys, and 4 users.

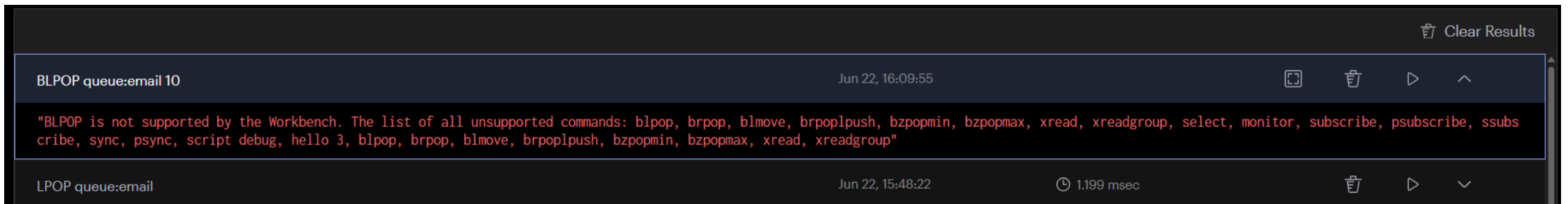
Clear Results				
LPOP queue:email	Jun 22, 15:48:22	1.199 msec		
{"to":"user@example.com","template":"welcome"}				
RPUSH queue:email '{"to":"user@example.com","template":"welcome"}'	Jun 22, 15:48:22	0.888 msec		
(integer) 1				

작업이 없을 때 계속 확인하는 대신 블로킹 명령을 사용할 수 있습니다.

```
BLPOP queue:email 10
```



BLPOP은 결과를 기다리는 블로킹 명령어라서 Workbench에서 지원하지 않습니다.



BLPOP vs LPOP 차이

	LPOP	BLPOP
데이터 없을 때	<code>nil</code> 즉시 반환	데이터 올 때까지 대기
용도	단순 조회	실시간 큐 처리
Workbench UI	지원	미지원